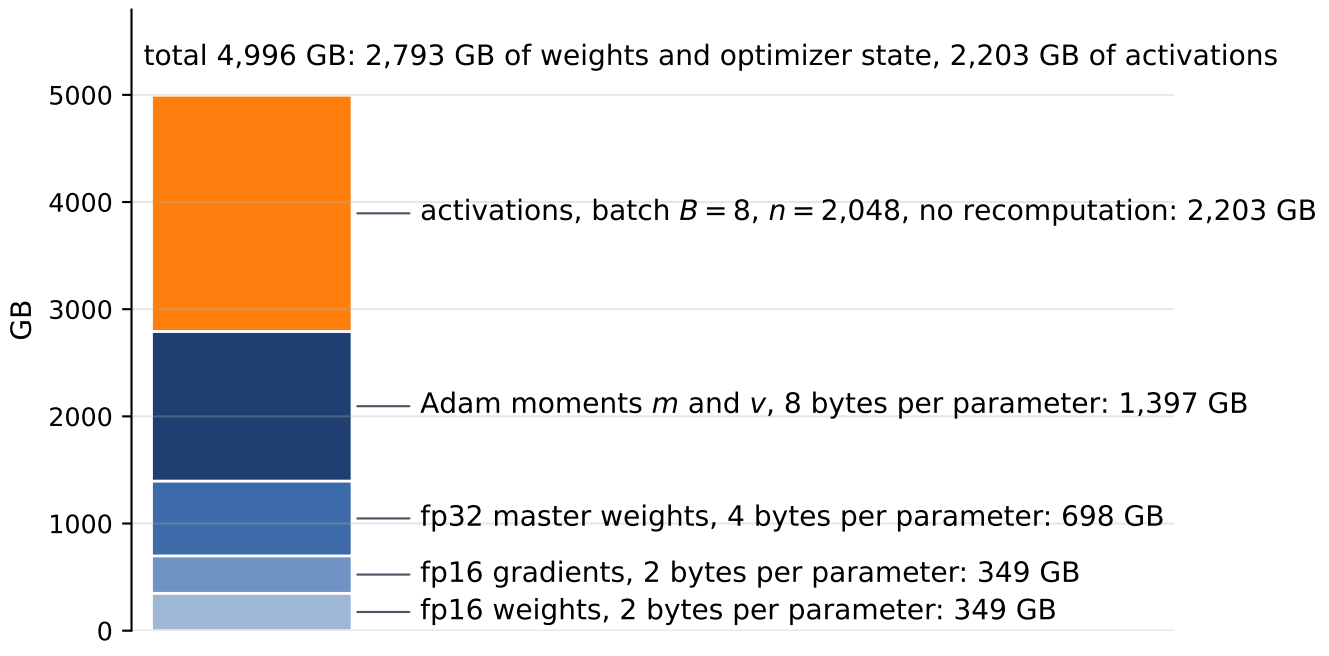
total 4,996 GB: 2,793 GB of weights and optimizer state, 2,203 GB of activations
5000
4000
3000
2000
1000
0
GB
activations, batch $B = 8$, $n = 2{,}048$, no recomputation: 2,203 GB
Adam moments m and v, 8 bytes per parameter: 1,397 GB
fp32 master weights, 4 bytes per parameter: 698 GB
fp16 gradients, 2 bytes per parameter: 349 GB
fp16 weights, 2 bytes per parameter: 349 GB